

Zhiling Zhou

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Education

Yale University

Aug 2025 – Present

PhD Student, Department of Statistics and Data Science

Fudan University

Sept 2020 – Jun 2025

B.S. in Data Science and Big Data Technology

School of Data Science, Shanghai, China

Publications and Preprints

1. How Transformers Represent Hierarchies: A Local-to-Global Mechanism.

Zhiling Zhou, Tianhao Wang, Zhuoran Yang.

ICML 2026.

2. Demystifying Classifier-Free Guidance for Auto-Regressive Image Generation.

Zhiling Zhou, Jiachun Pan, Fengzhuo Zhang, Dirk Bergemann, Zhuoran Yang.

Mechanistic Interpretability Workshop at ICML 2026.

3. Incremental Gauss–Newton Methods with Superlinear Convergence Rates.

Zhiling Zhou, Zhuanghua Liu, Chengchang Liu, Luo Luo.

Journal of Optimization Theory and Applications. [Paper Link](#)

4. DiffSim: Aligning Diffusion Model and Molecular Dynamics Simulation for Blind Docking.

Yanlin Fei, Yurui Li, Zhiling Zhou, Haohan Wang.

IEEE BIBM 2024. [Paper Link](#)

5. Robust Network Learning via Inverse Scale Variational Sparsification.

Zhiling Zhou, Zirui Liu, Chengming Xu, Yanwei Fu, Xinwei Sun.

Preprint. [arXiv:2409.18419](https://arxiv.org/abs/2409.18419)

Research Experience

How Transformers Represent Hierarchical Language Structure

Yale University

Advisors: Tianhao Wang, Zhuoran Yang

- Bridge Transformer-based context-free grammar generation with classical parsing algorithms.
- Develop a theoretical framework showing how Transformers implement parsing algorithm.

Demystifying Classifier-Free Guidance for Auto-Regressive Image Generation

Yale University

Advisors: Zhuoran Yang

- Demonstrated the repetition phenomenon and visual degradation in AR models lacking CFG.
- Theoretically modeled how image inductive bias shapes training dynamics and feature representation.

- Submitted to NeurIPS 2026. Preprint version to appear on arXiv.

Incremental Gauss–Newton Method for Nonlinear Equations

Fudan University

Advisor: Luo Luo

- Proposed an incremental Gauss–Newton method with superlinear convergence rates.
- Extended the method with mini-batch updates to improve convergence speed.

Robust Network Learning via Image Sparsification

Fudan University

Advisors: Xinwei Sun, Yanwei Fu

- Developed algorithms based on total variation and efficient Lasso optimization.
- Proposed a graph-based numerical acceleration technique.
- Improved robustness and interpretability of visual learning systems.

Aligning Diffusion Models and Simulations for Blind Docking

University of Illinois Urbana-Champaign

Advisor: Haohan Wang

- Aligned diffusion models with molecular dynamics simulations for blind docking.
- Constructed reliable blind docking datasets using molecular simulations.

Awards

- Third Prize of Excellent Student Scholarship, Fudan University (2020, 2022)
- KLA Scholarship, Fudan University & KLA Corporation (Top 4 undergraduates per year), 2024
- Outstanding Undergraduate Thesis Award, Fudan University, 2025.